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<https://github.com/madaossama/CCAS.4.3---Software-Engineering/tree/main>

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Functional and Non-Functional Requirements in Databases

Bank System

ID	Category	Functional Requirement	Justification
FR-M01	MUST	The system must store customer personal data (name, address, phone, national ID).	Needed to create a customer account and maintain identity records.
FR-M02	MUST	The system must allow the creation of bank accounts for customers (checking/savings).	Without account creation, the bank cannot operate.
FR-M03	MUST	The system must store account balance and update it after transactions.	Fundamental to banking operations.
FR-M04	MUST	The system must record every deposit transaction.	Required for tracking funds movement.
FR-M05	MUST	The system must record every withdrawal transaction.	Ensures financial traceability.
FR-M06	MUST	The system must store transaction history for each account.	Required for audits and customer inquiry.
FR-M07	MUST	The system must uniquely identify each account through an account number.	Prevents duplication/conflicts.
FR-M08	MUST	The system must authenticate users before allowing access.	Prevents unauthorized access to financial data.

ID	Category	Functional Requirement	Justification
FR-M09	MUST	The system must support fund transfer between accounts.	Core banking function.
FR-M10	MUST	The system must link each account to an owner/customer.	Establishes ownership and liability.
FR-S11	SHOULD	The system should support loan information storage (loan type, amount, interest).	Banking often includes loan services.
FR-S12	SHOULD	The system should store branch information (location, branch ID).	Required for multi-branch systems.
FR-S13	SHOULD	The system should allow account status changes (active, suspended, closed).	Allows proper lifecycle management.
FR-S14	SHOULD	The system should log failed login attempts.	Helps detect suspicious activities.
FR-S15	SHOULD	The system should allow role-based access for employees (teller, manager, admin).	Improves security and authorization control.
FR-S16	COULD	The system could generate monthly account statements.	Useful for customers and compliance.
FR-S17	COULD	The system could support saving customer relationship/contact history	Enhances CRM features.
FR-S18	COULD	The system could allow storing beneficiaries for faster transfers.	Improves user convenience.
FR-S19	COULD	The system could support scheduling recurring payments.	Extra feature for better service.
FR-W20	WONT HAVE	The system will not include investment portfolio management in this version.	Out of scope for current release, may be added later.

ID	Category	Non-Functional Requirement	Justification
NFR-M01	MUST	The database must ensure data consistency and ACID compliance.	Banking transactions require high accuracy.
NFR-M02	MUST	The database must encrypt stored sensitive data (e.g., passwords, national IDs).	Security and privacy protection is mandatory.
NFR-S03	SHOULD	The system should respond to common queries within ≤ 2 seconds.	Improves user experience and performance.
NFR-S04	SHOULD	The database should be scalable to support growth in number of customers and transactions.	Bank systems grow over time.
NFR-S05	SHOULD	The database should be available 99.9% of the time.	Banking services must be available almost always.